Rohan Surana

San Diego, CA | 510-265-4239 | rohansurana2810@gmail.com | linkedin.com/in/rohansurana28/ | https://rohan2810.github.io/

Summary

Researcher at Qualcomm working on proactive, personalized LLM agents and long-horizon context management and memory. Published research on RL post-training, preference optimization, and agent evaluation at NeurIPS, ICML, COLM, ICLR, and ACL.

Education

Masters of Science in Data Science — *March 2026*
University of California, San Diego (UCSD) GPA: 3.88/4.00. Advisor: Julian McAuley — San Diego, CA

Bachelor of Science in Software Engineering — *May 2022*
San Jose State University, GPA: 3.87/4.00 **(Summa cum laude)** — San Jose, CA

Experience

Qualcomm AI Research — San Diego, CA
Research Scientist — Apr 2026 – Present
- Study how to train proactive agents to decide when to ask users for clarification and what to ask during long-horizon tasks

Dell Technologies — Hopkinton, MA
AI Research Intern — June 2025 – Sept 2025
- Built multi-agent LLM system (LangGraph/LangChain + vLLM) with request batching and KV-cache, cutting p95 latency by 40%
- Designed scalable RAG pipelines integrating LLMs with vector databases using hybrid search and intelligent chunking strategies
- Engineered short/long-term memory with retrieval caching to reduce serving cost & exposed APIs for integration with existing workflows
- Developed agent-based monitoring using MCP and A2A protocols to automate telemetry collection, anomaly detection, and remediation

Dell Technologies — Santa Clara, CA
Software Engineer II — Mar 2024 – Aug 2024
- Architected TOSCA-based framework to orchestrate resources, cutting provisioning time 30% and improving infrastructure flexibility
- Built real-time infrastructure digital twin with predictive analytics using graph databases to enable drift detection and safe remediation

Software Engineer I — Jul 2022 – Mar 2024
- Accelerated cluster time-to-ready by 20% through automated Kubernetes operators supporting GPU workloads, KServe, and TorchServe
- Enhanced service performance by redesigning API and proto services with gRPC, leading a 4-intern team to boost efficiency 15%

Selected Publications (full list on Google Scholar)

- **Reasoning Pathologies in Large Language Models: A Diagnostic Perspective**
  Rohan Surana, J. Wu, S. Yu, G. Mundada, X. Jiang, Z. Huang, R. A. Rossi, J. McAuley, T. Yu — *NeurIPS 2026*
- **AMPS: Adaptive Modality Preference Steering via Functional Entropy**
  Z. Huang, X. Li, **Rohan Surana**, T. Yu, R. Wang, J. McAuley, J. Shang, J. Wu — *NeurIPS 2026*
- **WS-GRPO: Weakly-Supervised Group-Relative Policy Optimization for Rollout-Efficient Reasoning**
  G. Mundada*, Z. Huang*, **Rohan Surana***, S. Yu, J. Y. Zhang, X. Li, T. Yu, L. Yao, J. Shang, J. McAuley, J. Wu — *ICML 2026*
- **Evaluating Language Model Pluralism through In-the-wild Crowd Discussions**
  G. Mundada, **Rohan Surana**, N. Swaminathan, B. P. Majumder, J. Wu, J. McAuley, Z. Xie — *ACL 2026*
- **Importance Sampling for Multi-Negative Multimodal Direct Preference Optimization**
  X. Li, C. Wang, J. Wu, **Rohan Surana**, T. Yu, J. McAuley, J. Shang — *ICLR 2026*
- **In-context Ranking Preference Optimization**
  J. Wu*, **Rohan Surana***, Z. Xie, Y. Shen, Y. Xia, T. Yu, R. Rossi, P. Ammanabrolu, J. McAuley — *COLM 2025*
- **Image Difference Captioning via Adversarial Preference Optimization**
  Z. Huang, J. Wu, **Rohan Surana**, T. Yu, D. Arbour, R. Sinha, J. McAuley — *EMNLP 2025*
- **MASS-DPO: Multi-Negative Active Sample Selection for Direct Policy Optimization**
  Rohan Surana*, X. Li, S. Yu, Y. J. Shen, C. Wang, T. Yu, P. Ammanabrolu, J. Shang, J. McAuley, J. Wu — *Under Review*
- **F-GRPO: Factorized Group-Relative Policy Optimization for Unified Candidate Generation and Ranking**
  Rohan Surana, G. Mundada, J. Wu, X. Li, Y. Jiao, B. Jin, S. Zhou, T. Yu, R. Sinha, J. Han, J. Shang, J. McAuley — *Under Review*
- **Skill-R1: Agent Skill Evolution via Reinforcement Learning**
  Y. Vishe, **Rohan Surana**, X. Jiang, Z. Huang, X. Li, N. L. Kuang, T. Yu, R. A. Rossi, J. Shang, J. McAuley, J. Wu — *Under Review*
- **MusiCRS: Benchmarking Audio-Centric Conversational Recommendation**
  Rohan Surana*, A. Namburi*, G. Mundada*, A. Lal*, Z. Novack, J. McAuley, J. Wu — *ICASSP 2026*
- **An Empirical Study on Zero-Data Bootstrapping for Conversational Recommender Systems**
  Rohan Surana*, J. Wu*, Z. Xie*, Y. Xia, N. Kallus, J. McAuley — *EMNLP 2026 (Findings)*

Technical Skills

ML & Agents: PyTorch, Transformers, TensorFlow, scikit-learn, Hugging Face, vLLM, LangGraph
Engineering: Python, Go, Java, SQL, Kubernetes, Docker, FastAPI, gRPC, AWS, GCP